

Recall from last class:

- For a matrix A , define A_{ij} to be the matrix obtained from A by deleting row i and column j .
- For $n \geq 2$, the determinant of an $n \times n$ matrix $A = [a_{ij}]$ is

$$\det(A) = a_{11} \det(A_{11}) - a_{12} \det(A_{12}) + \cdots + (-1)^{n+1} a_{1n} \det(A_{1n}) = \sum_{j=1}^n (-1)^{1+j} a_{1j} \det(A_{1j}).$$

We call this the cofactor expansion across the first row.

- **Mohan's Theorem:** Cofactor expansion across any row or column is the same, and hence each cofactor expansion gives the determinant.
 - Let $A = [a_{ij}]$ be an $n \times n$ matrix, then A is **upper triangular** if $a_{ij} = 0$ when $i > j$.
- Theorem 3.2:** Let A be an $n \times n$ upper triangular matrix, then

$$\det(A) = a_{11} a_{22} \cdots a_{nn} = \prod_{j=1}^n a_{jj}.$$

- **Theorem 3.3:** If A is an $n \times n$ matrix and E is an elementary matrix, then

$$\det(EA) = \det(E) \det(A),$$

and

$$\det(E) = \begin{cases} 1 & \text{if } E \text{ represents adding a multiple of a row to another row,} \\ -1 & \text{if } E \text{ represents interchanging two rows,} \\ r & \text{if } E \text{ represents scaling a row by a factor of } r. \end{cases}$$

Theorem 3.6: If A and B are n by n matrices, then

$$\det(AB) = \det(A) \det(B).$$

1. Prove the theorem.

2. **Corollary:** A is invertible if and only if $\det(A) \neq 0$. Prove the corollary.

3. True or False:

(a) If A is an $n \times n$ invertible matrix, then $\det(A^{-1}) = \frac{1}{\det(A)}$.

(b) If A is an $n \times n$ matrix, then $\det(A) = \det(A^T)$.

(c) If $A \sim B$, then $\det(A) = \det(B)$.

4. We can think of $\det$ as a function from the set of $n \times n$ matrices to the real numbers. Is this function linear?

5. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given by $T(\vec{x}) = A\vec{x}$ where $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$. Let S_1 be the unit square given by $S_1 = \{(x, y) : 0 \leq x, y \leq 1\}$.

(a) Find $T(S_1)$.

(b) How does the area of S_1 related to the area of $T(S_1)$?

(c) Suppose $S_2 = \{(x, y) : 0 \leq x, y \leq 2\}$. How does the area of S_2 related to the area of $T(S_2)$?

6. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given by $T(\vec{x}) = A\vec{x}$ where $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Let S be the unit square given by $S = \{(x, y) : 0 \leq x, y \leq 1\}$.

(a) Find $T(S)$.

(b) How does the area of S related to the area of $T(S)$?

Theorem: If A is a 2×2 matrix, then the area of the parallelogram determined by the columns of A is $|\det A|$. If A is a 3×3 matrix, then the volume of the parallelepiped is $|\det A|$.

7. Find the area of the parallelogram with vertices $(-1, 0)$, $(0, 5)$, $(1, -4)$, and $(2, 1)$.

Theorem: Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be linear with associated matrix A . If S is a parallelogram in $\mathbb{R}^2$, then $\text{Area}(T(S)) = |\det A| \text{Area}(S)$.

We will generalize this theorem in Math 22b using calculus to more general functions!

Determinant Geometry – Answers / Solutions

1. *Proof.* We prove the result by cases.

Case 1: A is invertible.

Since A is invertible, the invertible matrix theorem tells us that there exists elementary row operations $E_1, \dots, E_p$ such that $I_n = E_p \dots E_1 A$. Since elementary matrices are invertible and the inverse of an elementary matrix is another elementary matrix, we get $A = E_1^{-1} \dots E_p^{-1}$. Thus we get $AB = E_1^{-1} \dots E_p^{-1} B$. Now we apply the previous theorem multiple times to get the following

$$\begin{aligned} \det AB &= \det E_1^{-1} \dots E_p^{-1} B \\ &= \det E_1^{-1} \det E_2^{-1} \dots \det E_p^{-1} \det B \\ &= \det E_1^{-1} \dots E_p^{-1} \det B \\ &= \det A \det B. \end{aligned}$$

Case 2: A is not invertible.

There must be a sequence of elementary matrices $E_1, \dots, E_p$ such that $A = E_1 \dots E_p U$ where U is the reduced row echelon form of A . Since A is not invertible, the invertible matrix theorem gives that $U \neq I_n$. Moreover, U must be upper triangular and have a zero along the diagonal. Thus $\det A = \det E_1 \dots E_p U = \det E_1 \dots \det E_p \det U = 0$ since $\det U = 0$.

Since A is not invertible, we know AB is not invertible. Thus we can apply the same reasoning to AB to conclude $\det AB = 0$. Hence $\det AB = 0 = \det A \det B$. $\square$

2. *Proof.* We prove the two implications.

$\implies$ Suppose A is invertible, then $AA^{-1} = I_n$. By the previous theorem, $1 = \det A \det A^{-1}$, and hence $\det A \neq 0$.

$\impliedby$ Suppose $\det A \neq 0$. We know there are elementary matrices $E_1, \dots, E_p$ such that $A = E_1 \dots E_p U$ where U is the reduced row echelon form of U . We know $\det E_k \neq 0$ for all $1 \leq k \leq p$. Thus $\det U \neq 0$. Since U is an upper triangular matrix, we know the determinant of U is the product of the diagonal elements. Thus U cannot have a zero along the diagonal. Thus U has all ones along the diagonal. Since U is a square matrix, we get $U = I_n$. By the invertible matrix theorem, A must be invertible. $\square$

3. (a) **True.** Since A is invertible, we know $AA^{-1} = I_n$. Thus $1 = \det I_n = \det AA^{-1} = \det A \det A^{-1}$, and hence $\det A^{-1} = \frac{1}{\det A}$.

- (b) **True.** Let a_{ij} be the entry in row i and column j of A and let b_{ij} be the entry in row i and column j of A^T .

Expand the determinant of A along the first column and use $a_{ij} = b_{ji}$; namely,

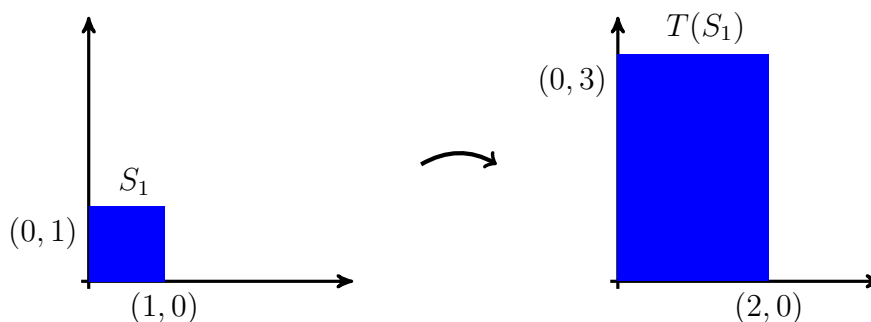
$$\begin{aligned} \det A &= a_{11} \det A_{11} - a_{21} \det A_{21} + \dots + (-1)^{n+1} a_{(n+1)1} \det A_{(n+1)1} \\ &= b_{11} \det(A^T)_{11} - b_{12} \det(A^T)_{12} + \dots + (-1)^{n+1} b_{1n} \det(A^T)_{1n} = \det A^T. \end{aligned}$$

- (c) **False.** Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, then $A \sim B$ and $\det A = 1$ and $\det B = -1$.

4. Let $M_{n \times n}$ be the set of $n \times n$ matrices, then $\det : M_{n \times n} \rightarrow \mathbb{R}$ is a function. This function fails to be linear since $\det I_2 = 1$ but $\det 2I_2 = 4$.

It is worth noting that if we instead defined the function $f_j : \mathbb{R}^n \rightarrow \mathbb{R}$ by $\det [\vec{a}_1 \dots, \vec{a}_{j-1} \ x \ \vec{a}_{j+1} \dots \vec{a}_n]$, then f_j is linear. Thus the determinant is linear in each column; we call this property **multi-linear**.

5. (a) In the figure below, we see how S_1 transforms under the transformation T .



(b) Notice that $\text{Area}(S_1) = 1$ and $\text{Area}(T(S_1)) = 6$.

(c) Here the area of S_2 is 4, and the area of $T(S_2)$ is 24, so we see that

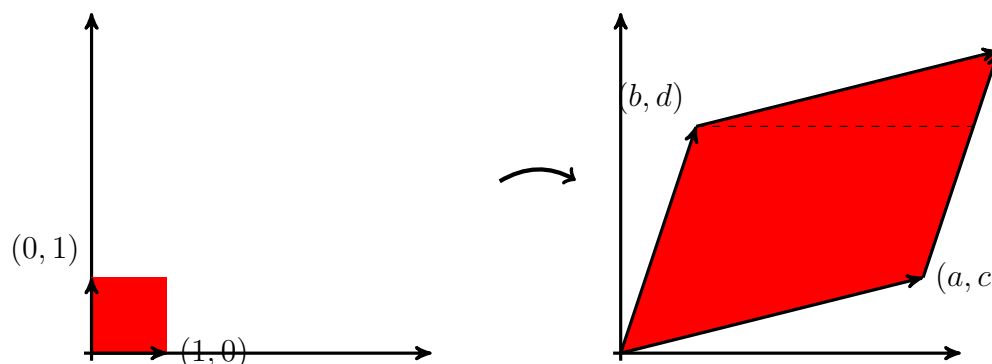
$$\text{Area}(T(S_2)) = 6 \cdot \text{Area}(S_2).$$

In this case, we see that T seems to scale the area of 6, which also happens to be the determinant of the associated matrix. Thus it seems reasonable to guess that

$$\text{Area}(T(S_2)) = |\det A| \cdot \text{Area}(S_2)$$

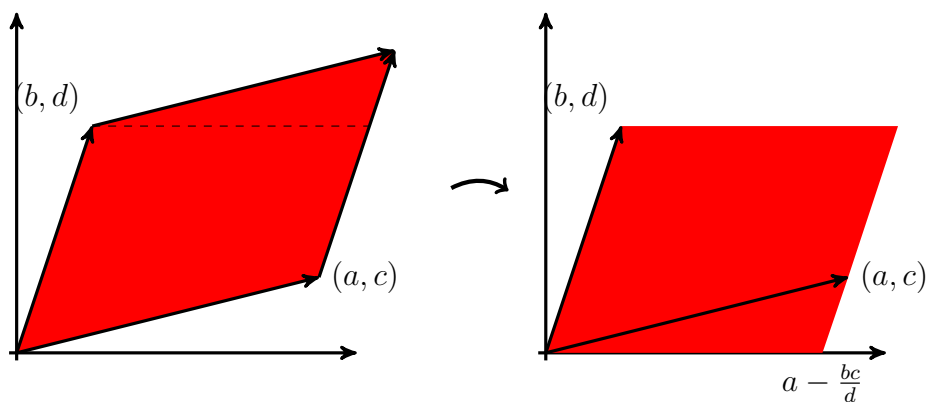
more generally.

6. (a) We show $T(S)$ in the following figure.

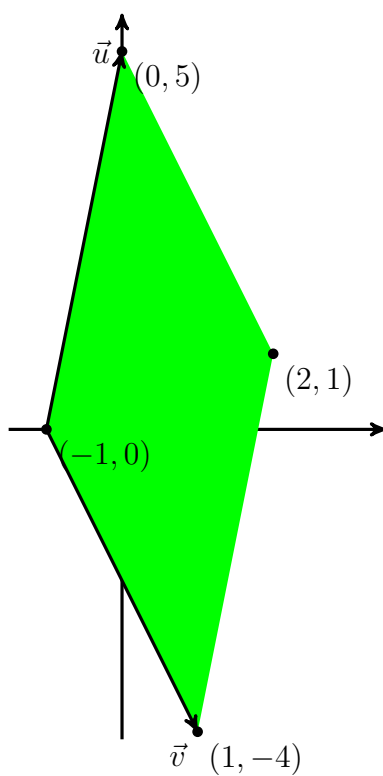


(b) In the next figure, we show how to rearrange the parallelogram to compute the area. In the parallelogram on the right, we see that the area is given by $d(a - \frac{bc}{d}) = ad - bc = \det A$. Thus we see that

$$\text{Area}(T(S)) = \det A.$$



7. We plot the given points below.



We define $\vec{u}$ and $\vec{v}$ to represent adjacent sides of the parallelogram; namely, let $\vec{u} = \begin{bmatrix} 0 \\ 5 \end{bmatrix} - \begin{bmatrix} -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 1 \\ -4 \end{bmatrix} - \begin{bmatrix} -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \end{bmatrix}$. Thus the area of parallelogram is given by $\left| \det \begin{bmatrix} 1 & 2 \\ 5 & -4 \end{bmatrix} \right| = |-14| = 14$.